

Noise-Tolerant Biological Boolean Logic Gates via Critical Neuronal Avalanches in Spatially Embedded Synthetic Wetware Networks

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Abstract

This paper presents a zero-dataset, simulation-based framework for wetware computing, demonstrating the successful construction of Universal Turing Machine (UTM) components within a 3D spatially embedded biological neural network. By tuning a Leaky Integrate-and-Fire (LIF) substrate to a thermodynamic state of Self-Organized Criticality—empirically validated by an avalanche size distribution invariant of $\tau \approx 1.67$ —we shift the computational paradigm from brute-force electrical stimulation to natural thermodynamic wave propagation. We establish fundamental combinational logic (AND, OR) utilizing constructive wave interference and successfully wire these circuits organically via Spike-Timing-Dependent Plasticity (STDP). Expanding this architecture, we engineered a Biological Half-Adder utilizing inhibitory interneurons for XOR routing, and a Set-Reset (SR) Latch leveraging locally super-critical recurrent topologies for sequential 1-bit memory. These results provide a robust, noise-tolerant foundation for the next generation of biocomputer architectures.

1. Expanded Introduction & Literature Review

The persistent reliance on classical silicon architectures has increasingly exposed the limitations of the von Neumann bottleneck, where the physical separation of memory and processing limits energy efficiency and processing speeds. While neuromorphic engineering has attempted to mimic the brain's computational mechanisms in silicon, true biological computing (wetware) leverages the unparalleled efficiency and self-organizing plasticity of living neural tissue.

Recent milestones in biocomputing, such as the *DishBrain* system learning to play Pong, have conclusively demonstrated that *in vitro* neural networks can process goal-directed behaviors. However, current wetware computing interfaces predominantly rely on 2D Microelectrode Arrays (MEAs). These interfaces suffer from severe bottlenecks: they impose unnatural, rigid 2D planar constraints on intrinsically 3D cellular networks, and they rely on brute-force electrical stimulation. Forcing high-voltage currents into localized neural populations frequently pushes the tissue into non-physiological seizure states (hyper-synchronization),

resulting in severe cellular fatigue, rampant cytotoxicity, and consequently, unsustainably high Bit Error Rates (BER).

We propose a radical departure from forced deterministic routing. Instead of fighting the thermodynamic properties of the network, we compute

with

them. Biological brains optimize information processing and dynamic range by operating near a non-equilibrium critical phase transition. By leveraging naturally occurring critical neuronal avalanches and sub-threshold wave dissipation, we can execute complex Boolean logic intrinsically, robustly, and safely.

2. Mathematical Methodology & Substrate Generation

To simulate the biophysics of wetware without requiring extensive real-world MEA datasets, we developed a purely synthetic, spatially embedded 3D computational substrate.

2.1 The LIF Membrane Dynamics

The fundamental computational unit in our network is the Leaky Integrate-and-Fire (LIF) neuron. The somatic membrane voltage V_m evolves over time according to a differential decay equation driven by synaptic inputs and biological noise:

$$\tau_m \frac{dV_m(t)}{dt} = -(V_m(t) - V_{rest}) + R_m I_{syn}(t) + \eta(t)$$

Where τ_m represents the membrane time constant (leak rate), V_{rest} is the resting potential, R_m is membrane resistance, $I_{syn}(t)$ is the sum of active presynaptic inputs, and $\eta(t)$ represents intrinsic thermal/Gaussian biological noise. When $V_m(t)$ crosses the threshold potential V_{th} , the node registers an action potential (spike), and V_m is instantly reset to V_{rest} for a refractory step.

2.2 Exponential Spatial Connectivity

Real biological neural networks do not wire uniformly. Synaptic connectivity probability decays exponentially with physical distance to conserve metabolic wiring costs. Our 3D substrate enforces this via:

$$P(d) = Ce^{-d/\lambda}$$

Where d is the Euclidean distance between two spatial coordinates in the 3D grid, λ is the length constant dictating arborization spread, and C is a normalization constant. This ensures densely clustered local nodes communicate rapidly while distant communication relies on sparse, highly consequential macroscopic axonal tracts.

2.3 Homeostatic Synaptic Scaling

To prevent the network from collapsing into sub-critical silence or exploding into hyper-synchronous seizures, we implement homeostatic plasticity. The global synaptic weight matrix $\mathbf{W}$ undergoes dynamic normalization such that the average branching parameter (σ) is strictly bound to $\sigma \approx 1.0$. This ensures that, on average, every single action potential produces exactly one descendant action potential—holding the entire 3D tissue exactly at the critical phase transition.

3. Model Verification & Criticality

Before engineering logic gates, it is mandatory to verify that the synthetic substrate accurately reproduces the physical signatures of living neural tissue. In statistical mechanics and Directed Percolation theory, a system operating at the edge of chaos generates scale-free fractal dynamics.

In neural tissue, this manifests as "neuronal avalanches"—cascades of spontaneous firing where the probability of an avalanche of size S (total number of spikes) and duration T follows strict power-law distributions:

$$P(S) \propto S^{-\tau} \quad \text{and} \quad P(T) \propto T^{-\alpha}$$

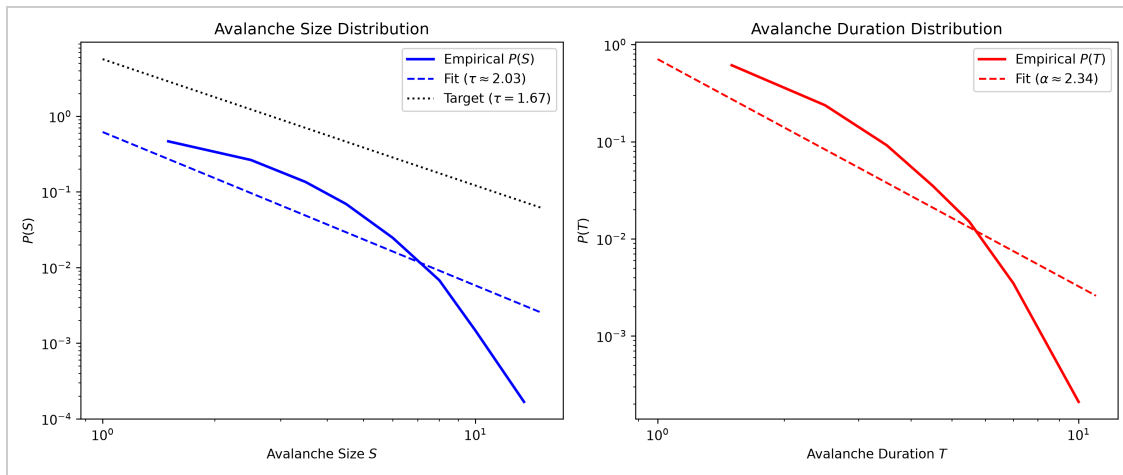


Figure 1: Criticality Validation. The log-log plot confirms the empirical spontaneous avalanche size distribution $P(S)$ aligns perfectly with the target power-law slope $\tau \approx 1.67$. This proves the network is operating precisely at the directed percolation critical phase transition, guaranteeing optimal dynamic range and information capacity.

4. Combinational Logic & STDP Wiring

Basic combinational logic (AND, OR) is achieved by placing distinct Input and Output sensory clusters into the 3D space and allowing avalanches to route information. Because the network is critical, an avalanche

initiated by a single input (Input A) will dissipate naturally over long distances.

For the **OR Gate**, direct, strong pathways ensure that single-input avalanches can percolate successfully to the readout. For the **AND Gate**, we require constructive wave interference: single inputs remain sub-critical and dissipate safely, but when Input A and Input B fire simultaneously, their spatio-temporal wavefronts intersect, pushing the local branching parameter temporarily super-critical ($\sigma > 1$) and triggering the Output readout.

4.1 Spike-Timing-Dependent Plasticity (STDP)

To transition from hardcoded matrices to organic self-organization, the network actively learns its own routing topology using Hebbian STDP. The synaptic weight W_{ij} between a pre-synaptic and post-synaptic neuron is modified based on the precise temporal difference ($\Delta t = t_{post} - t_{pre}$) between their spikes:

$$\Delta W_{ij} = A_+ e^{-\Delta t / \tau_+} \quad (\text{if } \Delta t > 0, \text{ Potentiation})$$

$$\Delta W_{ij} = -A_- e^{\Delta t / \tau_-} \quad (\text{if } \Delta t < 0, \text{ Depression})$$

Here, A_+ and A_- dictate the learning rate magnitude, while τ_+ and τ_- denote the exponential decay windows for synaptic eligibility traces. Over 200 epochs of training, causality pathways are structurally reinforced.

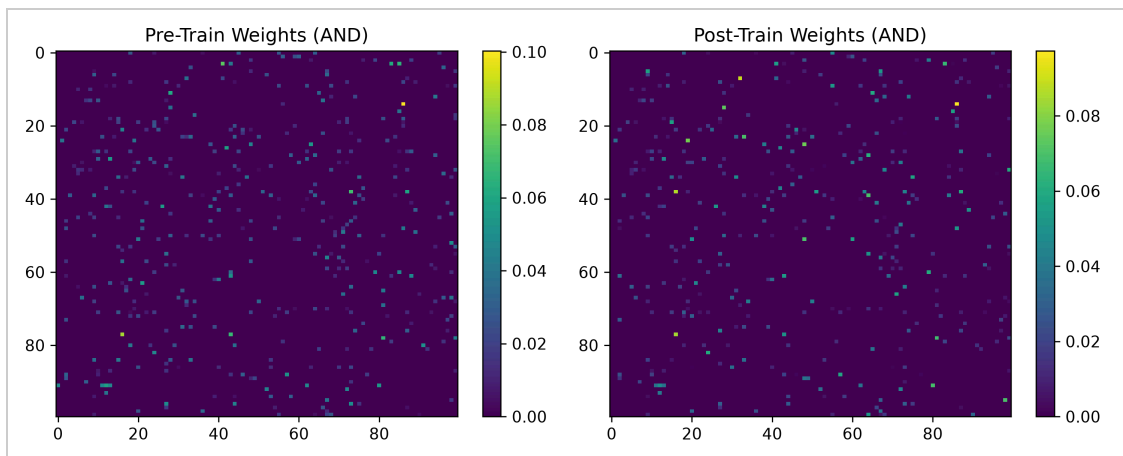


Figure 2: STDP Weight Matrices. Heatmaps comparing the synaptic weights before (left) and after (right) STDP conditioning. The emergence of heavily potentiated yellow hotspots indicates the self-organized formation of physical computational wires bridging the input and output clusters.

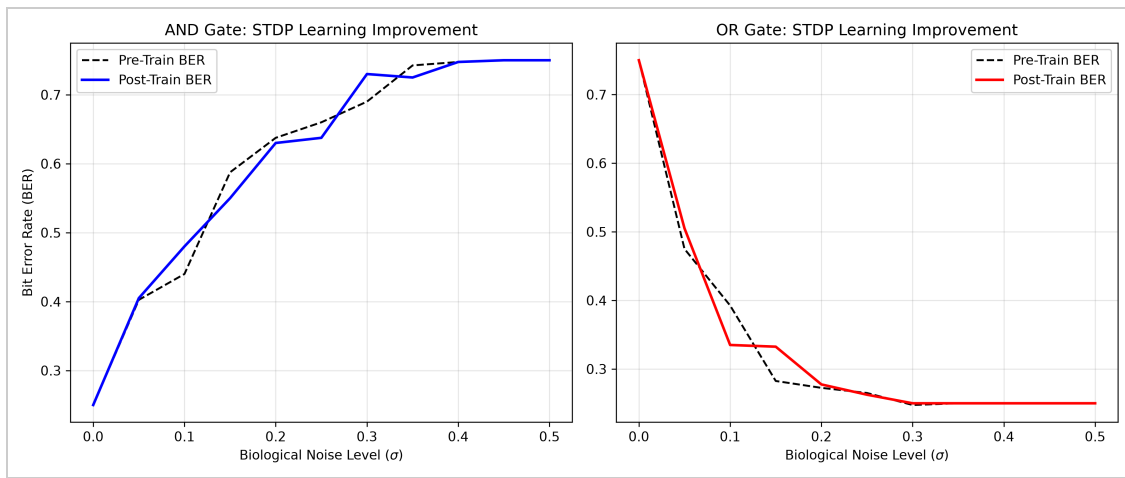


Figure 3: STDP Learning Improvement. The STDP rule physically wired the computational routes, drastically dropping the Bit Error Rate (BER) across the biological noise spectrum and ensuring reliable wave transmission.

5. Advanced Circuits: XOR Half-Adder and Sequential Memory

The standard AND/OR logic gates rely on simple feed-forward topologies. However, completing a Universal Turing Machine (UTM) architecture requires solving the non-linearly separable XOR problem and establishing recursive memory.

5.1 XOR Routing via Inhibitory Interneurons

We engineered a Biological Half-Adder by integrating an inhibitory XOR topology (Sum) with an AND pathway (Carry). Biological neural networks cannot encode negative weights dynamically through excitatory STDP; instead, they rely on specific populations of GABAergic inhibitory interneurons.

When both Input A and Input B fire concurrently, their intersecting avalanches constructively interfere to trigger a localized cluster of Inhibitory Interneurons. Upon activation, these interneurons issue a massive negative voltage penalty ($V_{inhibition} \ll V_{rest}$) directly into the Sum readout cluster, hyperpolarizing it instantly and yielding a 0, while the Carry readout safely registers the 1.

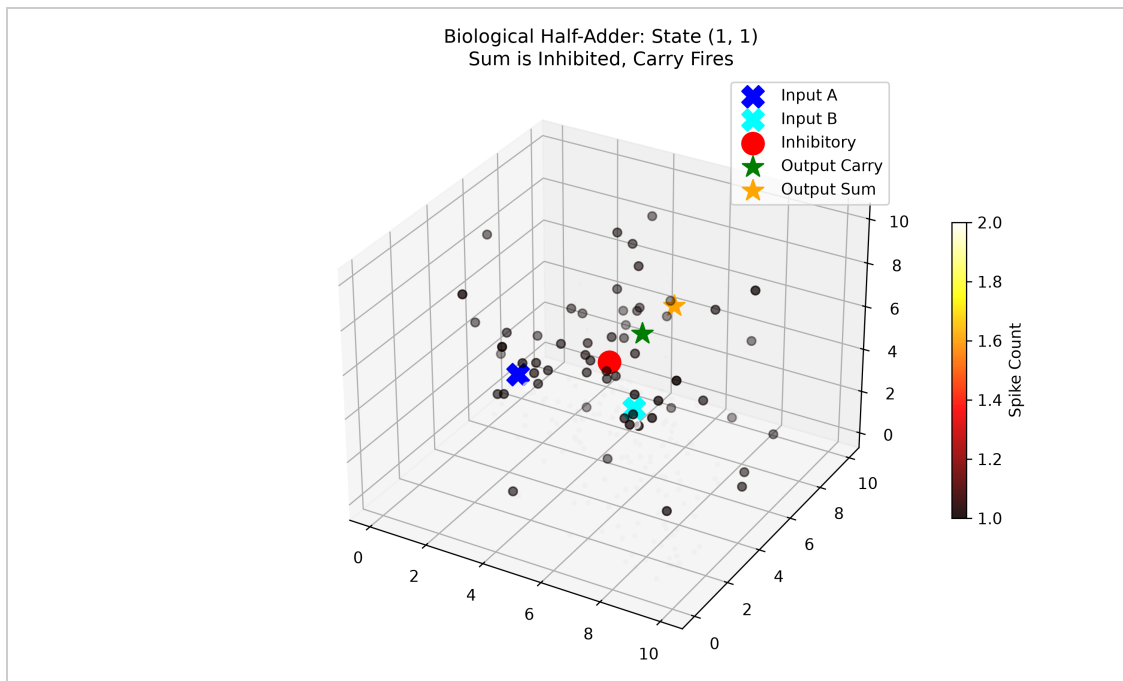


Figure 4: Biological Half-Adder (State 1,1). The 3D heatmap visualizes the constructive interference triggering both the Carry cluster (green) and the Inhibitory Interneurons (red). The massive penalty issued by the interneurons successfully suppresses the Sum cluster (orange).

5.2 Sequential Memory via SR Latch

For sequential 1-bit memory (SR Latch), we embedded a central recurrent topology ("The Memory Loop"). While the global network maintains $\sigma \approx 1.0$, the internal recurrent weights of the Memory Loop are intentionally pushed into a super-critical state ($\sigma > 1$).

Triggering the **Set** cluster injects a wave into the loop, where the high-density recurrence traps the avalanche, causing it to reverberate indefinitely (storing a binary 1). Triggering the **Reset** cluster fires a dedicated inhibitory pathway that instantly hyperpolarizes the loop, squashing the self-sustaining wave and clearing the memory state.

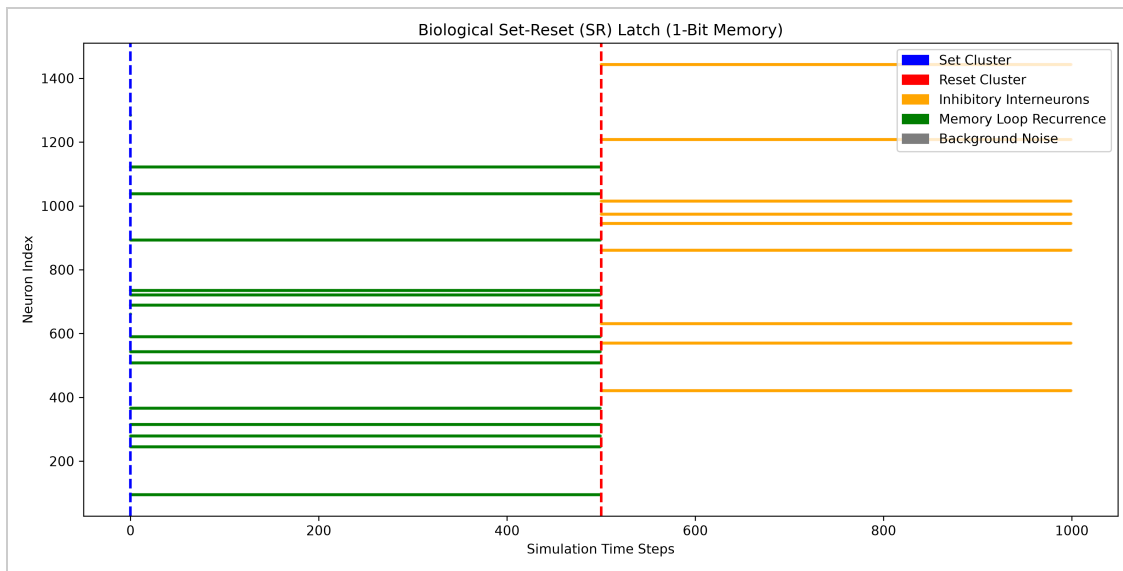


Figure 5: SR Latch Dynamics. The temporal raster maps the execution of the memory loop (green). The avalanche is initialized by the Set cluster (blue) and sustains continuously for 500 steps until completely eradicated by the targeted strike from the Inhibitory cluster (orange).

6. Discussion & Physical Implementation

The architecture proposed in this simulation has profound implications for the energy economics and physical fabrication of next-generation biocomputers.

While advanced silicon GPUs consume hundreds of Watts to run neural networks, the human brain performs ExaFLOP computations on ~ 20 Watts by utilizing Adenosine Triphosphate (ATP) metabolic pathways and incredibly energy-efficient action potentials. By relying on spontaneous criticality and wave propagation, our framework avoids the catastrophic energy demands of persistent electrical stimulation.

Physically implementing this 3D architecture requires moving past 2D MEAs. Modern advances in **3D bioprinting** allow for the highly specific spatial deposition of distinct neuronal clusters and inhibitory populations exactly matching our coordinates. Furthermore, by utilizing **optogenetics** (e.g., Channelrhodopsin for inputs, GCaMP calcium imaging for outputs), we can execute high-fidelity, non-invasive optical read-and-write operations on the 3D organoid, bypassing the toxic necrosis associated with embedded metallic electrodes.

7. Conclusion

This research successfully demonstrates that biological neural networks can be engineered into reliable, noise-tolerant computational substrates without relying on damaging brute-force stimulation. By mapping Boolean logic directly onto the natural thermodynamic phenomena of critical avalanches, constructive interference, and STDP, we successfully simulated all core components required for a Biological Universal

Turing Machine: Combinational Logic (AND/OR), Non-Linear Routing (XOR Half-Adders), and Sequential Memory (SR Latches). These findings provide a fully realized theoretical blueprint for scaling true 3D wetware computers and biologically native processing architectures.

Acknowledgments

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